M2 ICFP
Theoretical Condensed Matter

B. Douçot, B.Estienne, L. Messio

Problem: The Jordan Wigner transformation

At first glance spin models may seem more difficult to solve than fermionic or bosonic ones, as the commutation relations of spin operators are more involved than those of bosonic or fermionic operators. Thus, it is quite tempting to map a spin model to a fermionic or bosonic model. While this is straightforward for an isolated spin 1/2 (see Sec. 1), it is more tricky when several spins are involved. Jordan and Wigner proposed such a transformation with fermions on a 1D chain (see Sec. 2). The resulting model can then be solved using the bosonization technique (see Sec. 3).

The spin vector operator on site i is $\mathbf{S}_i$, of components S_i^x, S_i^y and S_i^z. A spin 1/2 state is represented by a two dimensional vector (we chose z as the quantization axis, as is usually done, and denote by $|\uparrow\rangle$, $|\downarrow\rangle$ the corresponding basis) and the spin component operators are given by the Pauli matrices $S_i^\alpha = \frac{1}{2}\sigma_\alpha$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{1}$$

We define the S^+ and S^- operators that increase or decrease S^z by one:

$$\begin{aligned} S^+ &= S_x + iS_y, \\ S^- &= S_x - iS_y. \end{aligned}$$

1 Mapping for a single site

1. We define the fermionic operator $f^\dagger = S^+$. The idea is that a site has only two states both in spin and fermionic representation : it is either occupied/up ($f^\dagger|0\rangle \leftrightarrow |\uparrow\rangle$) or empty/down ($|0\rangle \leftrightarrow |\downarrow\rangle$).

   What is the expression of S_x, S_y in terms of fermionic operators ? Deduce the expression of S_z from S_x and S_y.

2. Are the spin commutation relations respected ? What is the anticommutator $\{S^+, S^-\}$? Is it compatible with the fermionic expression ?

2 Mapping for a 1D chain

So far we have a mapping between spin operators and fermionic operators, *on a single site*. Spin and fermionic operators $\mathbf{S}_i$ and $f_i^\dagger$ now have a site index i. We consider an antiferromagnetic periodic chain with N atoms of spin 1/2 with nearest neighbor anisotropic interactions:

$$H = J\sum_{i=0}^{N-1}(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z). \tag{2}$$

J is a positive energy and $\Delta \geq 0$ characterizes the anisotropy (for $\Delta = 1$, we recover the SU(2) symmetric Heisenberg Hamiltonian $H = J\sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}$). We assume periodic boundary conditions ($\mathbf{S}_N = \mathbf{S}_0$).

3. What is the issue with the previous mapping when several sites are involved ?

4. We chose a basis of states that are eigenvectors of all the S_i^z. A basis spin state with p up spins is labelled by their positions $i_1, i_2, \dots i_p$ (with $i_j < i_{j+1}$): $|i_1, i_2, \dots, i_p\rangle = S_{i_1}^+ \dots S_{i_p}^+ | \downarrow\downarrow \dots \downarrow\rangle$. It corresponds to a fermionic state with p particles where the i_j sites are filled: $f_{i_1}^\dagger f_{i_2}^\dagger \dots f_{i_p}^\dagger |0\rangle$ (up to a phase factor that we fix to 1 for ordered i_j's). Show that

$$S_j^+ |i_1, i_2, \dots, i_p\rangle \leftrightarrow (-1)^k f_j^\dagger f_{i_1}^\dagger f_{i_2}^\dagger \dots f_{i_p}^\dagger |0\rangle,$$

where k is the integer such that $i_k \leq j < i_{k+1}$. We want to obtain an expression of S_j^+ in terms of fermionic operators leading to correct anticommutation and commutation relations. Check that the following mapping does the trick:

$$S_j^+ = e^{i\pi \sum_{p=0}^{j-1} n_p} f_j^\dagger = (-1)^{\sum_{p=0}^{j-1} n_p} f_j^\dagger.$$

5. The exponential operator is known as a string operator. Check that its presence does no spoil the on-site commutation relations $[S_i^a, S_i^b] = i\epsilon^{abc} S_i^c$.

6. Check that spin operators (expressed in terms of fermionic operators) at different sites now commute, as they should.

7. We suppose that $S_{tot}^z = \sum_i S_i^z$ is conserved (it is the case for the Hamiltonian Eq. (2)). Thus, we can work in the subspace of states with fixed S_{tot}^z. What does it imply for the fermionic states ? What boundary condition must we impose on the fermionic operators to recover the periodic boundary conditions for the spins in such a subspace ?

8. Rewrite the anisotropic Heisenberg Hamiltonian of Eq. (2) in terms of fermionic operators. Be cautious for the term $S_j^\alpha S_{j+1}^\alpha$ for $j = N - 1$.

9. In which case can we exactly and easily solve it ?

10. We suppose from now that the number of sites N is even ($N = 2N'$) and that $\Delta = 0$. Show that the eigenstates are Slater determinants of plane waves. What are then the allowed wave vectors ?

11. Show that for each value of p, the ground state is unique. What is the value of p that minimizes the energy ? What is the corresponding S^z value ? What do you think about that ?

12. What are the correlations $\langle S_j^z S_0^z \rangle$ (density-density) for $J_z = 0$ and $j \neq 0$ in the ground state ? What are they in the thermodynamic limit, for fixed j ?

    We would now like to calculate the correlation functions in the xy plane: $\langle S_m^x S_n^x + S_m^y S_n^y \rangle$. Up to a factor 1/2, this is equal to $\langle S_m^+ S_n^- + S_m^- S_n^+ \rangle$.

13. We suppose that $m < n$. We have:

$$S_m^+ S_n^- = f_m^\dagger f_n e^{i\pi \sum_{m \leq j < n} n_j}.$$

    Start by calculating $\langle f_m^\dagger f_n \rangle$ in the ground state.

    The difficulty to calculate $\langle S_m^+ S_n^- + S_m^- S_n^+ \rangle$ comes from the string operator $e^{i\pi \sum_{m \leq j < n} n_j}$. Below, we will see how to calculate it in the thermodynamic limit, using bosonization.

3 Bosonization

Bosonization is a mapping between system of interacting fermions in (1+1) dimensions and a system of massless, non-interacting bosons. While bosonization is an approximation in finite size, it becomes exact in the continuum (*i.e.* at low energies and long wave-lengths).

3.1 Continuum limit

Before taking the thermodynamic limit, we first re-introduce the lattice spacing a, so that sites are located at positions $x = ja$, and the total chain length is $L = Na$. Correspondingly the momentum is now $Q = \frac{k}{a}$, where k is the previous momentum for $a = 1$. The thermodynamic limit is obtained by taking $a \to 0$ (or equivalently $N \to \infty$), keeping L constant. The relevant momenta values in this limit are $Q = \pm\frac{\pi}{2a} + q$ with (for even N):

$$q = \frac{2\pi}{L}\left(n - \frac{1}{2}\right), \qquad n \in \mathbb{Z}. \tag{4}$$

(depending on the parity of the number of particle in the ground state $p = N/2$, the boundary conditions are either periodic or antiperiodic, but near $\pm\frac{\pi}{2a}$, the previous formula is always valid). Accordingly the fermion operator f_j now reads

$$f_j = \frac{1}{\sqrt{N}}\sum_k \tilde{f}_k e^{ikj} \to \sqrt{\frac{a}{L}}\left(e^{i\frac{\pi}{2}j}\sum_q \tilde{f}_{\frac{\pi}{2}+aq}e^{iqx} + e^{-i\frac{\pi}{2}j}\sum_q \tilde{f}_{-\frac{\pi}{2}+aq}e^{iqx}\right), \qquad \text{where } x = ja$$

Introducing the left and right moving fermion as

$$\Psi_L(x) = \sqrt{\frac{1}{L}}\sum_q \underbrace{\tilde{f}_{\frac{\pi}{2}+aq}}_{c_L(q)} e^{iqx}, \qquad \Psi_R(x) = \sqrt{\frac{1}{L}}\sum_q \underbrace{\tilde{f}_{-\frac{\pi}{2}+aq}}_{c_R(q)} e^{iqx} \tag{5}$$

we get

$$\boxed{f_j = \sqrt{a}\left(e^{i\pi j/2}\Psi_L(x) + e^{-i\pi j/2}\Psi_R(x)\right), \qquad x = ja} \tag{6}$$

14. Argue that in the limit $a \to 0$ the Fermi momentum becomes infinite, and as a consequence the two fermions Ψ_L and Ψ_R become independent : the lattice fermion operator f_j yields two fermions in the continuum ! This phenomenon is due to the fact that there are two momentum region in the low-energy limit. Comment on the phase factors in (6). Show that we have

$$\boxed{\{\Psi^\dagger_\eta(x), \Psi_{\eta'}(x')\} = \delta_{\eta,\eta'}\delta(x - x'), \qquad \Psi_\eta(x + L) = -\Psi_\eta(x), \qquad \eta, \eta' \in \{L, R\}.} \tag{7}$$

15. Show that in the continuum limit, the non-interacting fermionic Hamiltonian

$$H = J\sum_{j=0}^{N-1}\left(\frac{f_j^\dagger f_{j+1} + f_{j+1}^\dagger f_j}{2}\right)$$

becomes

$$H = v_F \int_0^L dx\, i\left(\Psi_L^\dagger(x)\partial_x\Psi_L(x) - \Psi_R^\dagger(x)\partial_x\Psi_R(x)\right) = v_F\sum_q q\left(c_R^\dagger(q)c_R(q) - c_L^\dagger(q)c_L(q)\right) \tag{8}$$

What is the Fermi velocity v_F ?

16. Let us denote by $|0\rangle$ the ground state (Fermi see) of the continuum model, which we are going to call the *vacuum*. Show that for $q > 0$ we have

$$c_R^\dagger(-q)|0\rangle = 0 \qquad \text{and} \qquad c_R(q)|0\rangle = 0 \tag{9}$$

$$c_L^\dagger(q)|0\rangle = 0 \qquad \text{and} \qquad c_L(-q)|0\rangle = 0 \tag{10}$$

Comment on the vacuum energy.

3.2 Normal ordering

The vacuum energy problem is typical of a field theory, and is due to the infinite number of degrees of freedom. It is however very easy to circumvent. Before taking the continuum limit, we can substract the GS energy from the Hamiltonian, which amounts to put the GS energy to 0. The Hamiltonian in the continuum becomes

$$\boxed{H_F = v_F \sum_q q\left(: c_R^\dagger(q) c_R(q) : - : c_L^\dagger(q)_L(q) :\right)} \tag{11}$$

where the normal ordered product $: c_\eta^\dagger(k) c_\eta(k) :$ is defined as

$$: c_\eta^\dagger(k) c_\eta(k) := c_\eta^\dagger(k) c_\eta(k) - \langle 0| c_\eta^\dagger(k) c_\eta(k) |0\rangle$$

17. Show that

$$: c_L^\dagger(k) c_L(k) := \begin{cases} -c_L(k) c_L^\dagger(k) & \text{if} \quad k > 0 \\ c_L^\dagger(k) c_L(k) & \text{if} \quad k < 0 \end{cases}, \quad : c_R^\dagger(k) c_R(k) := \begin{cases} c_R^\dagger(k) c_R(k) & \text{if} \quad k > 0 \\ -c_R(k) c_R^\dagger(k) & \text{if} \quad k < 0 \end{cases} \tag{12}$$

Electronic density : Since the Fermi see contains an infinite number of fermions in the continuum limit, the electronic density $\Psi_\eta^\dagger(x)\Psi_\eta(x)$ is not well defined (the zero mode, which counts the total number of fermions, is infinite). To obtain a sensible quantity in the thermodynamic limit, we have to subtract this infinite quantity. Again, the proper way to do this is to work with the *normal ordered* quantities $\rho_\eta(x) =: \Psi_\eta^\dagger(x)\Psi_\eta(x) :$ defined as

$$: \Psi_\eta^\dagger(x)\Psi_\eta(x) := \Psi_\eta^\dagger(x)\Psi_\eta(x) - \langle 0|\Psi_\eta^\dagger(x)\Psi_\eta(x)|0\rangle \tag{13}$$

Upon decomposing the density in Fourier modes

$$\rho_\eta(x) = \frac{1}{L}\sum_q \rho_\eta(q) e^{-iqx} \tag{14}$$

we get the usual $\rho_\eta(q) = \sum_k c_\eta^\dagger(k+q) c_\eta(k)$, except for $q = 0$. The regularized zero mode $\rho_\eta(q = 0) = N_\eta$ is

$$\rho_\eta(q = 0) = \sum_k \underbrace{\left(c_\eta^\dagger(k) c_\eta(k) - \langle 0| c_\eta^\dagger(k) c_\eta(k) |0\rangle\right)}_{:c_\eta^\dagger(k) c_\eta(k):} = \frac{1}{L}\hat{N}_\eta$$

Show that

$$\hat{N}_L = L\left(\sum_{k<0} c_L^\dagger(k) c_L(k) - \sum_{k>0} c_L(k) c_L^\dagger(k)\right)$$

$$\hat{N}_R = L\left(-\sum_{k<0} c_R(k) c_R^\dagger(k) + \sum_{k>0} c_R^\dagger(k) c_R(k)\right)$$

Note that this is now finite, since $\hat{N}_\eta$ counts the deviation in fermion number (of type η) from the reference state $|0\rangle$.

18. Show that (for $q, q' \neq 0$)

$$[\rho_\eta(q), \rho_{\eta'}(q')] = \delta_{q+q',0}\delta_{\eta,\eta'} \sum_k \left(\langle 0| c_\eta^\dagger(k+q) c_\eta(k+q) |0\rangle - \langle 0| c_\eta^\dagger(k) c_\eta(k) |0\rangle\right) \tag{15}$$

$$= \delta_{q+q',0}\delta_{\eta,\eta'} \begin{cases} -\frac{qL}{2\pi} & \text{if} \quad \eta = R \\ \frac{qL}{2\pi} & \text{if} \quad \eta = L \end{cases} \tag{16}$$

3.3 Bosonic field

We now introduce a right and left (compact) scalar field $\phi(x)$ and $\bar{\phi}(x)$ as

$$\phi(x) = \phi_0 - \frac{2\pi x}{L} a_0 + i \sum_{n>0} \frac{1}{n} \left(a_n e^{i2\pi nx/L} - a_n^\dagger e^{-i2\pi nx/L} \right) \tag{17}$$

$$\bar{\phi}(x) = \bar{\phi}_0 + \frac{2\pi x}{L} \bar{a}_0 + i \sum_{n>0} \frac{1}{n} \left(\bar{a}_n e^{-i2\pi nx/L} - \bar{a}_n^\dagger e^{i2\pi nx/L} \right) \tag{18}$$

with commutation relations

$$[a_n, a_m^\dagger] = n\delta_{n,m}, \qquad [\bar{a}_n, \bar{a}_m^\dagger] = n\delta_{n,m}, \qquad [\phi_0, a_0] = [\bar{\phi}_0, \bar{a}_0] = i \tag{19}$$

all other commutators vanishing.

19. Check that the following identifications are compatible with these commutation relations

$$a_n = \rho_R(-q), \qquad a_n^\dagger = \rho_R(q), \qquad a_0 = N_R$$
$$\bar{a}_n = \rho_L(q), \qquad \bar{a}_n^\dagger = \rho_L(-q), \qquad \bar{a}_0 = N_L$$

where $q = \frac{2\pi n}{L} > 0$. This amounts to identify

$$\boxed{\rho_R(x) = -\frac{1}{2\pi} \partial_x \phi(x), \qquad \rho_L(x) = \frac{1}{2\pi} \partial_x \bar{\phi}(x)} \tag{20}$$

20. Out of these bosonic modes we define so-called *vertex operators* as

$$V_\alpha(x) = \left(\frac{2\pi}{L} \right)^{\frac{\alpha^2}{2}} e^{-i\frac{2\pi x}{L}\frac{\alpha^2}{2}} : e^{i\alpha\phi(x)} : \tag{21}$$

where the normal ordered exponential is

$$: e^{i\alpha\phi(x)} : = e^{i\alpha\phi_0} \exp\left[\alpha \sum_{n>0} \frac{1}{n} a_n^\dagger e^{-iqx} \right] \exp\left[-\alpha \sum_{n>0} \frac{1}{n} a_n e^{iqx} \right] e^{-i\alpha \frac{2\pi x}{L} a_0} \tag{22}$$

where $q = 2\pi n/L$. Likewise we define $\bar{V}_\alpha = \left(\frac{2\pi}{L}\right)^{\frac{\alpha^2}{2}} e^{i\frac{2\pi x}{L}\frac{\alpha^2}{2}} : e^{i\alpha\bar{\phi}(x)} :$.

The Campbell-Baker-Hausdorff (CBH) formula states that $e^A e^B = e^{A+B} e^{\frac{1}{2}[A,B]}$ provided A and B commute with $[A, B]$. Show that the product of two vertex operators is

$$: e^{i\alpha\phi(x)} :: e^{i\alpha'\phi(x')} := : e^{i\alpha\phi(x) + i\alpha\phi(x')} : \left(e^{-i\frac{2\pi x}{L}} - e^{-i\frac{2\pi x'}{L}} \right)^{\alpha\alpha'} . \tag{23}$$

21. Deduce that

$$\langle 0 | V_\alpha(x) V_{-\alpha}(x') | 0 \rangle = \left(i\frac{L}{\pi} \sin\frac{\pi}{L}(x' - x) \right)^{-\alpha^2} \tag{24}$$

22. The bosonized expressions of the fermion operators are

$$\Psi_R(x) = \frac{1}{\sqrt{2\pi}} V_{-1}(x), \qquad \Psi_R^\dagger(x) = \frac{1}{\sqrt{2\pi}} V_1(x)$$
$$\Psi_L(x) = \frac{1}{\sqrt{2\pi}} \bar{V}_1(x), \qquad \Psi_L^\dagger(x) = \frac{1}{\sqrt{2\pi}} \bar{V}_{-1}(x)$$

Recompute $\langle 0 | f_m^\dagger f_n | 0 \rangle$ using bosonization.

23. Show that the string operator becomes local in the boson, namely

$$\boxed{\sum_{j=m}^{n-1} f_j^\dagger f_j = \frac{n-m}{2} + \frac{1}{2\pi}\left(\bar{\phi}(x') - \phi(x')\right) - \frac{1}{2\pi}\left(\bar{\phi}(x) - \phi(x)\right)} \tag{25}$$

where $x = ma$ and $x' = na$.

24. In order to have a hermitian expression in the continuum limit, we use a slightly modified expression of the string operator

$$\cos(\pi \sum_{j=m}^{n-1} f_j^\dagger f_j) \tag{26}$$

In the continuum limit this modified string operator become

$$\frac{1}{2}\left[e^{i\pi(n-m)/2} e^{-\frac{i}{2}\left(\bar{\phi}(x)-\phi(x)\right)} e^{\frac{i}{2}\left(\bar{\phi}(x')-\phi(x')\right)} + h.c.\right] \tag{27}$$

Show that for large distances the correlation function $\langle S_m^+ S_n^- \rangle$ decay algebraically as

$$\langle S_m^+ S_n^- \rangle \sim \frac{1}{(n-m)^{\frac{1}{2}}} \tag{28}$$